Marco Sèvegrand

Software Engineer | Cybersecurity & Secure Systems

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Website LinkedIn GitHub HackTheBox Credly

About Me

Software Engineering graduate now pursuing a Master's in Cybersecurity. Focused on application security, applied cryptography, secure software development lifecycles, and threat modeling. Solid foundation in infrastructure, networking, and systems architecture. AWS Certified AI Practitioner and Cloud Practitioner. Comfortable with Go, C, and Python.

Experience

Full-Stack Web Developer Feb 2026 – Present

Biobrassica Braga, Portugal

- Sole developer for the company's online presence: built a full branding page and e-commerce store with product CRUD, inventory management, and secure payment integration via IfThenPay.
- Initially developed with Django/Python on a Linux VPS; currently migrating the stack to Laravel/PHP on cPanel to simplify maintenance and handover.

Store Associate Feb 2026 – Present

Biobrassica Braga, Portugal

- Oversaw stock management and replenishment, ensuring consistent product availability.
- Delivered front-line customer service, handling inquiries and in-store sales.

Math & Physics Tutor Sep 2025 – Present

Self-employed Remote

- Mentored high-school students in mathematics and physics through one-to-one lessons.
- Designed study plans aligned to each student's exam goals and skill level.

Author — "Mistério no Aeroclube" 2019

- ↗ Published fiction book (80 pages, teen demographic).

Certifications

AWS Certified AI Practitioner (AIF-C01)

Amazon Web Services (AWS) · July 2026

AWS Certified Cloud Practitioner (CLF-C02)

Amazon Web Services (AWS) · July 2026

English C1 Advanced

Cambridge Assessment English · May 2023

AWS Badges in: Cloud Essentials, AI-DLC

Education

B.Sc. Software Engineering Sep 2023 – Jun 2026

University of Minho Braga, Portugal

– Final grade: 16.15/20. Coursework includes distributed systems, cybersecurity, operating systems, compilers, algorithms, databases, AI, and software systems design.

Secondary Education Sep 2020 – Jun 2023

Escola Secundária Fernão de Magalhães Braga, Portugal

– Final grade: 19.6/20. Participated in Mathematics Olympiads, Chemistry Olympiads, and Youth Parliament.

Skills

Programming	Go, Python, Bash.
Systems	Linux, Docker, VPS.
Security	Applied Cryptography, Threat Modeling, Secure SDLC, Certificate-based Auth.
Architecture	Concurrent Programming, Network Protocols, Distributed Systems.
Learning	AWS, SOC Analysis/Ops, Kubernetes.
Soft Skills	Proactive, clear communication, active listening, critical thinking, learning agility.
Spoken	Portuguese (Native), English (C1).

Selected Projects

Secure Multi-Device Messaging Platform Grade: 18/20, B+

Python, TCP/TLS, E2EE, X3DH, Ed25519, X25519, AES-GCM, Argon2id, SQLite, Textual ↗ GitHub

- Built an end-to-end encrypted messaging system with multi-device identity, offline delivery, group messaging, local history, and a terminal TUI.
- Implemented X3DH session bootstrap, symmetric ratcheting, device challenge-response authentication, certificate-based identity binding, and at-rest encryption for client/server secrets.

Automated Software Testing & Quality Analysis Pipeline Grade: 18.5/20, A-

Java, Maven, JUnit 5, jqwik, JaCoCo, PIT, EvoSuite, Make ↗ GitHub

- Strengthened two Java applications with unit tests, property-based tests, coverage analysis, mutation testing, and automated test generation.
- Automated the full quality pipeline with Maven/Make, expanding the suite to 444 tests and achieving PIT mutation scores of 76% and 85% across the two projects.

Document Indexing and Search Service Grade: 20/20, A+

C, Linux, POSIX IPC, FIFOs, LRU Cache, Concurrency ↗ GitHub

- Built a Linux document indexing service around FIFO-based client-server communication with persistent indexing for stateful retrieval.
- Added an LRU cache and parallel keyword search, with process cleanup and concurrency control for safe multi-document execution.

Space Communication Protocols & Simulator Grade: 19.5/20, A+

Go, TCP/UDP, React, Node.js, CORE Emulator, FEC ↗ GitHub

- Designed custom TCP and UDP protocols with FEC, fragmentation, ordered delivery, and exponential backoff retransmission.
- Simulated a mothership-rover mission architecture and built a real-time dashboard validated in CORE under packet loss and latency.

LLM-Assisted Software Development Lifecycle Grade: 19/20, A

TypeScript, React, Fastify, PostgreSQL, Prisma, Zod, Turborepo, LLM-Augmented Engineering ↗ GitHub

- Conducted a complete software development lifecycle from requirements through architecture, implementation, testing, and documentation, using structured LLM assistance from Claude, GitHub Copilot, and Stitch AI; catalogued and critically reviewed all 48 prompts.
- Applied a three-pillar methodology of upfront planning, strict modular decomposition, and mandatory human review on every LLM output, explicitly distinguishing the approach from unstructured “vibe coding.”
- Selected among the top 4 projects in the curricular unit and featured at the ES3A workshop, held as part of Universidade do Minho’s Departamento de Informática 50th-anniversary celebrations.

Distributed Time-Series Storage System Grade: 18/20, B+

Java, TCP, Multithreading, Thread Pool, Pub/Sub, Persistence ↗ GitHub

- Built a concurrent multi-client TCP server with a custom framed protocol, thread-pool execution, and safe shared-state handling.
- Developed a configurable time-series storage engine with in-memory caching, disk persistence, and real-time pub/sub notifications.